



Matrix differentiation

- Function is a vector and the variable is a scalar

$$\mathbf{f}(t) = [f_1(t), f_2(t), \dots, f_n(t)]^T$$

Definition

$$\frac{d\mathbf{f}}{dt} = \left[\frac{df_1(t)}{dt}, \frac{df_2(t)}{dt}, \dots, \frac{df_n(t)}{dt} \right]^T$$



Matrix differentiation

- Function is a matrix and the variable is a scalar

$$F(t) = \begin{bmatrix} f_{11}(t) & f_{12}(t), \dots, f_{1m}(t) \\ f_{21}(t) & f_{22}(t), \dots, f_{2m}(t) \\ \vdots & \\ f_{n1}(t) & f_{n2}(t), \dots, f_{nm}(t) \end{bmatrix} = [f_{ij}(t)]_{n \times m}$$

Definition

$$\frac{dF}{dt} = \begin{bmatrix} \frac{df_{11}(t)}{dt} & \frac{df_{12}(t)}{dt}, \dots, \frac{df_{1m}(t)}{dt} \\ \frac{df_{21}(t)}{dt} & \frac{df_{22}(t)}{dt}, \dots, \frac{df_{2m}(t)}{dt} \\ \vdots & \\ \frac{df_{n1}(t)}{dt} & \frac{df_{n2}(t)}{dt}, \dots, \frac{df_{nm}(t)}{dt} \end{bmatrix} = \left[\frac{df_{ij}(t)}{dt} \right]_{n \times m}$$



Matrix differentiation

- Function is a scalar and the variable is a vector

$$f(\mathbf{x}), \mathbf{x} = (x_1, x_2, \dots, x_n)^T$$

Definition

$$\frac{df}{d\mathbf{x}} = \left[\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right]^T$$

In a similar way,

$$f(\mathbf{x}), \mathbf{x} = (x_1, x_2, \dots, x_n)$$

$$\frac{df}{d\mathbf{x}} = \left[\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right]$$



Matrix differentiation

- Function is a vector and the variable is a vector

$$\mathbf{x} = [x_1, x_2, \dots, x_n]^T, \mathbf{y} = [y_1(\mathbf{x}), y_2(\mathbf{x}), \dots, y_m(\mathbf{x})]^T$$

Definition

$$\frac{d\mathbf{y}}{d\mathbf{x}^T} = \begin{bmatrix} \frac{\partial y_1(\mathbf{x})}{\partial x_1}, \frac{\partial y_1(\mathbf{x})}{\partial x_2}, \dots, \frac{\partial y_1(\mathbf{x})}{\partial x_n} \\ \frac{\partial y_2(\mathbf{x})}{\partial x_1}, \frac{\partial y_2(\mathbf{x})}{\partial x_2}, \dots, \frac{\partial y_2(\mathbf{x})}{\partial x_n} \\ \vdots \\ \frac{\partial y_m(\mathbf{x})}{\partial x_1}, \frac{\partial y_m(\mathbf{x})}{\partial x_2}, \dots, \frac{\partial y_m(\mathbf{x})}{\partial x_n} \end{bmatrix}_{m \times n}$$



Matrix differentiation

- Function is a vector and the variable is a vector

$$\mathbf{x} = [x_1, x_2, \dots, x_n]^T, \mathbf{y} = [y_1(\mathbf{x}), y_2(\mathbf{x}), \dots, y_m(\mathbf{x})]^T$$

In a similar way,

$$\frac{d\mathbf{y}^T}{d\mathbf{x}} = \begin{bmatrix} \frac{\partial y_1(\mathbf{x})}{\partial x_1}, \frac{\partial y_2(\mathbf{x})}{\partial x_1}, \dots, \frac{\partial y_m(\mathbf{x})}{\partial x_1} \\ \frac{\partial y_1(\mathbf{x})}{\partial x_2}, \frac{\partial y_2(\mathbf{x})}{\partial x_2}, \dots, \frac{\partial y_m(\mathbf{x})}{\partial x_2} \\ \vdots \\ \frac{\partial y_1(\mathbf{x})}{\partial x_n}, \frac{\partial y_2(\mathbf{x})}{\partial x_n}, \dots, \frac{\partial y_m(\mathbf{x})}{\partial x_n} \end{bmatrix}_{n \times m}$$



Matrix differentiation

- Function is a vector and the variable is a vector

Example:

$$\mathbf{y} = \begin{bmatrix} y_1(\mathbf{x}) \\ y_2(\mathbf{x}) \end{bmatrix}, \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, y_1(\mathbf{x}) = x_1^2 - x_2, y_2(\mathbf{x}) = x_3^2 + 3x_2$$

$$\frac{d\mathbf{y}^T}{d\mathbf{x}} = \begin{bmatrix} \frac{\partial y_1(\mathbf{x})}{\partial x_1} & \frac{\partial y_2(\mathbf{x})}{\partial x_1} \\ \frac{\partial y_1(\mathbf{x})}{\partial x_2} & \frac{\partial y_2(\mathbf{x})}{\partial x_2} \\ \frac{\partial y_1(\mathbf{x})}{\partial x_3} & \frac{\partial y_2(\mathbf{x})}{\partial x_3} \end{bmatrix} = \begin{bmatrix} 2x_1 & 0 \\ -1 & 3 \\ 0 & 2x_3 \end{bmatrix}$$



Matrix differentiation

- Function is a scalar and the variable is a matrix

$$f(\mathbf{X}), \mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & & & \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix} \in \mathbb{R}^{m \times n}$$

Definition

$$\frac{df(\mathbf{X})}{d\mathbf{X}} = \begin{bmatrix} \frac{\partial f}{\partial x_{11}} & \frac{\partial f}{\partial x_{12}} & \cdots & \frac{\partial f}{\partial x_{1n}} \\ \cdots & & & \\ \frac{\partial f}{\partial x_{m1}} & \frac{\partial f}{\partial x_{m2}} & \cdots & \frac{\partial f}{\partial x_{mn}} \end{bmatrix}$$



Matrix differentiation

- Useful results

(1)

$$\mathbf{x}, \mathbf{a} \in \mathbb{R}^{n \times 1}$$

Then,

$$\frac{d\mathbf{a}^T \mathbf{x}}{d\mathbf{x}} = \mathbf{a}, \frac{d\mathbf{x}^T \mathbf{a}}{d\mathbf{x}} = \mathbf{a}$$



How to prove?



Matrix differentiation

- Useful results

$$(2) \quad \mathbf{x} \in \mathbb{R}^{n \times 1} \quad \text{Then,} \quad \frac{d\mathbf{x}^T \mathbf{x}}{d\mathbf{x}} = 2\mathbf{x}$$

$$(3) \quad \mathbf{y}(\mathbf{x}) \in \mathbb{R}^{m \times 1}, \quad \mathbf{x} \in \mathbb{R}^{n \times 1}, \quad \frac{d\mathbf{y}^T(\mathbf{x})}{d\mathbf{x}} = \left(\frac{d\mathbf{y}(\mathbf{x})}{d\mathbf{x}^T} \right)^T$$

$$(4) \quad A \in \mathbb{R}^{m \times n}, \quad \mathbf{x} \in \mathbb{R}^{n \times 1} \quad \text{Then,} \quad \frac{dA\mathbf{x}}{d\mathbf{x}^T} = A$$

$$(5) \quad A \in \mathbb{R}^{m \times n}, \quad \mathbf{x} \in \mathbb{R}^{n \times 1} \quad \text{Then,} \quad \frac{d\mathbf{x}^T A^T}{d\mathbf{x}} = A^T$$

$$(6) \quad A \in \mathbb{R}^{n \times n}, \quad \mathbf{x} \in \mathbb{R}^{n \times 1} \quad \text{Then,} \quad \frac{d\mathbf{x}^T A \mathbf{x}}{d\mathbf{x}} = (A + A^T)\mathbf{x}$$

$$(7) \quad \mathbf{X} \in \mathbb{R}^{m \times n}, \quad \mathbf{a} \in \mathbb{R}^{m \times 1}, \quad \mathbf{b} \in \mathbb{R}^{n \times 1} \quad \text{Then,} \quad \frac{d\mathbf{a}^T \mathbf{X} \mathbf{b}}{d\mathbf{X}} = \mathbf{a} \mathbf{b}^T$$



Matrix differentiation

- Useful results

$$(8) \quad \mathbf{X} \in \mathbb{R}^{n \times m}, \mathbf{a} \in \mathbb{R}^{m \times 1}, \mathbf{b} \in \mathbb{R}^{n \times 1} \quad \text{Then, } \frac{d\mathbf{a}^T \mathbf{X}^T \mathbf{b}}{d\mathbf{X}} = \mathbf{b} \mathbf{a}^T$$

$$(9) \quad \mathbf{X} \in \mathbb{R}^{m \times n}, \mathbf{B} \in \mathbb{R}^{n \times m} \quad \text{Then, } \frac{d(\text{tr} \mathbf{X} \mathbf{B})}{d\mathbf{X}} = \mathbf{B}^T$$

$$(10) \quad \mathbf{X} \in \mathbb{R}^{n \times n}, \mathbf{X} \text{ is invertible, } \frac{d|\mathbf{X}|}{d\mathbf{X}} = |\mathbf{X}| (\mathbf{X}^{-1})^T$$